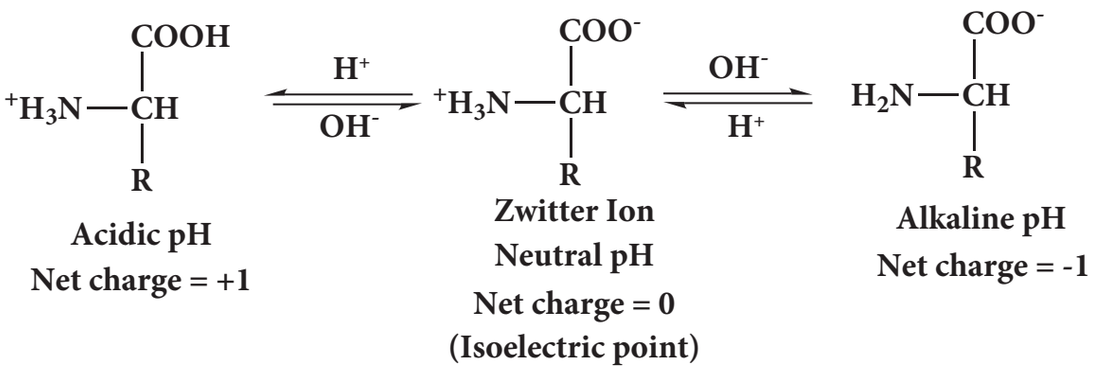
Amino acids can also be classified as essential and non-essential amino acids based on the ability to be synthesise by the human. The amino acids that can be synthesised by us are called non-essential amino acids (Gly, Ala, Glu, Asp, Gln, Asn, Ser, Cys, Tyr & Pro) and those needs to be obtained through diet are called essential amino acids (Phe, Val, Thr, Trp, Ile, Met, His, Arg, Leu and Lys). These ten essential amino acids can be memorised by mnemonic called PVT TIM HALL.

Although the vast majority of plant and animal proteins are formed by these 20 α- amino acids, many other amino acids are also found in the cells. These amino acids are called as non–protein amino acids. Example: ornithine and citrulline (components of urea cycle where ammonia is converted into urea)

14.2.3 Properties of amino acid

Amino acids are colourless, water soluble crystalline solids. Since they have both carboxyl group and amino group their properties differ from regular amines and carboxylic acids. The carboxyl group can lose a proton and become negatively charged or the amino group can accept a proton to become positively charged depending upon the pH of the solution. At a specific pH the net charge of an amino acid is neutral and this pH is called **isoelectric point**. At a pH above the isoelectric point the amino acid will be negatively charged and positively charged at pH values below the isoelectric point.

In aqueous solution the proton from carboxyl group can be transferred to the amino group of an amino acid leaving these groups with opposite charges. Despite having both positive and negative charges this molecule is neutral and has amphoteric behaviour. These ions are called zwitter ions.



Except glycine all other amino acids have at least one chiral carbon atom and hence are optically active. They exist in two forms namely D and L amino acids. However, L-amino acids are used predominantly by the living organism for synthesising proteins. Presence of D-amino acids has been observed rarely in certain organisms.

14.2.4 Peptide bond formation

The amino acids are linked covalently by peptide bonds. The carboxyl group of the first amino acid react with the amino group of the second amino acid to give an amide linkage between these amino acids. This amide linkage is called **peptide bond**. The resulting compound is called a dipeptide. Addition an another amino acid to this dipeptide a second peptide bond results in tripeptide. Thus we can generate tetra peptide, penta peptide etc… When you have

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